

v4.4 — Nottingham Grading (MedGemma 1.5)

E2E outcome: the system assesses tubule formation and nuclear pleomorphism from hotspot patches via MedGemma 1.5, combines them with the v4.3 mitotic score, and computes the Nottingham grade **in code**. It also proposes a histologic type requiring mandatory confirmation. Pathologist sees evidence per sub-score and can override anything; overrides recompute the grade and are watermarked downstream.

Est: 1.5 weeks. Depends on v4.3 (mitotic score confirmed).

1. Scope

In: patch sampling, MedGemma structured calls (tubule %, pleomorphism, histologic type), aggregation in code, deterministic grade synthesis, override UI with evidence panels. **Out:** HoVer-Net morphometry corroboration (v4.6 optional); any numeric computed by the LLM (hard rule from v3).

2. Patch sampling (`worker/grading.py`)

- From confirmed hotspots: sample `grading.n_patches` = 24 patches, 512² px @ 1.0 µm/px, normalized, stratified: rank all 10× grid patches (from v4.2) inside hotspots by tumor probability, take stratified draws across the top-50% (seeded RNG = slide checksum) — avoids grading only the single hottest patch (tumors are heterogeneous; pleomorphism should reflect the worst area per convention, so also force-include the top-3 probability patches).
- Persist the sampled patch set (ids + PNGs) — the evidence panel shows exactly what the model saw.

3. MedGemma calls

All calls: Vertex endpoint, temperature 0, max 2 schema retries then flag `needs_human`, prompt templates versioned in `configs/prompts/{name}@v1.md`, template sha in `model_versions`.

3.1 Per-patch tubule assessment (24 calls, parallel ≤ 4)

Prompt (abridged; full text in `configs/prompts/tubule@v1.md`):

Image: one 512×512 μm H&E region of invasive breast carcinoma. Estimate the percentage of tumor area in THIS patch forming clear glandular/tubular structures (lumen surrounded by polarized epithelium). Respond as JSON.

Schema (enforced via constrained decoding / `response_schema`):

```
{"tubule_percent": <int 0-100>, "tumor_present": <bool>, "confidence": <"low"|"medium"|"high">}
```

3.2 Per-patch pleomorphism (24 calls)

Assess nuclear pleomorphism of invasive tumor cells vs normal epithelium: size variation, chromatin texture, nucleolar prominence, vesicular change. Nottingham criteria: 1 = uniform, small; 2 = moderate variability; 3 = marked variation / vesicular / prominent nucleoli.

```
{"pleomorphism_score": <1|2|3>, "rationale": "<≤50 words>", "confidence": ...}
```

3.3 Histologic type (1 call, top-8 patches as multi-image input)

```
{"type": "<IDC-NST|ILC|mucinous|tubular|papillary|metaplastic|other>",
```

```
"differential": ["..."], "rationale": "<≤80 words>", "confidence": ...}
```

4. Aggregation + grade (`pipeline/grading.py` — pure, zero LLM)

```
def aggregate(tubule_responses, pleo_responses, mitotic_score, cfg):
```

```
    t = [r for r in tubule_responses if r.tumor_present]
```

```
    tubule_percent = weighted_median([r.tubule_percent for r in t],
```

```
                                     w=[CONF_W[r.confidence] for r in t]) # {low:.5, med:1, high:1.5}
```

```
    tubule_score = 1 if tubule_percent > 75 else (2 if tubule_percent >= 10 else 3)
```

```
    pleo_score = weighted_mode([r.pleomorphism_score ...]); ties → max(tied) # conservative: worse grade
```

```
    nottingham_sum = tubule_score + pleo_score + mitotic_score
```

```
    grade = 1 if nottingham_sum <= 5 else (2 if nottingham_sum <= 7 else 3)
```

```
flags = []
```

```
if len(t) < cfg.min_tumor_patches (=8):      flags += ["insufficient_tumor_patches"]
```

```
if disagreement(pleo_responses) > cfg.max_disp: flags += ["pleo_high_variance"] # e.g.  
>30% off-mode
```

```
return Grading(..., flags)
```

Any flag $\Rightarrow$ UI banner "low agreement — review carefully" and the affected sub-score rendered amber. Cutoffs (75/10, 3–5/6–7/8–9) live in `configs/scoring.yaml`, not code.

MedGemma additionally writes a findings narrative — one call, input = the final aggregated JSON only (grounded), output = plain paragraph for the review screen. It cannot alter numbers because it only ever receives them.

5. Review UI

Three sub-score cards + type card + grade card:

- **Tubule card:** score, derived %, histogram of per-patch estimates, click-through to each patch (viewer jump + crop). Override: direct score select (1/2/3) + optional corrected %; requires 10-char min justification text.
- **Pleomorphism card:** score, per-patch votes, rationales visible per patch. Same override affordance.
- **Mitotic card:** read-only summary from v4.3 (count, mm², score) + "reopen stage" link (re-enters mitosis review; grading re-runs on re-confirm).
- **Histologic type card:** proposed type + differential + rationale. **Blocked state by design** — neither "confirm stage" nor report generation proceeds until pathologist clicks Confirm type or edits it (decision #3). Type list = CAP value set in `configs/cap_elements.yaml`.
- **Grade card:** live $T+P+M = \text{sum} \rightarrow \text{Grade}$, recomputed on any override; overridden components show a "manually assigned" chip that will persist into the report.

Overrides $\rightarrow$ `review_edits` diff + `score_override` audit events `{component, from, to, justification}`.

6. Output JSON

```
{ "patches": [{"id": "p_017", "uri": "...", "tubule": {...}, "pleo": {...}],
```

```
"aggregate": {"tubule_percent": 22.0, "tubule_score": 2,
```

```

    "pleo_score": 3, "mitotic_score": 3,

    "nottingham_sum": 8, "grade": 3, "flags": []},

    "histologic_type": {"type": "IDC-NST", "differential": ["ILC"], "rationale": "..."},

    "narrative": "...",

    "model_versions": {"medgemma": "1.5@...", "prompts": {"tubule": "v1", "pleo": "v1", "type": "v1"}} }

```

DB: **gradings** row written at confirm (post-override values + **overrides** jsonb + machine values for delta analytics).

```

CREATE TABLE gradings (

    case_id uuid PRIMARY KEY REFERENCES cases(id),

    tubule_percent real, tubule_score int, pleo_score int,

    mitotic_score int, nottingham_sum int, grade int,

    histologic_type text NOT NULL, type_confirmed_by text NOT NULL,

    machine jsonb NOT NULL, overrides jsonb NOT NULL DEFAULT '{}');

```

7. Tests

- **grading.py** aggregation: table-driven — weighted median/mode incl. tie → worse-grade rule; boundary %s (exactly 75, exactly 10); flag triggers.
- Grade function: all 27 sub-score combinations → expected grade (exhaustive).
- Schema-violation path: fake client returns malformed JSON twice → third failure flags **needs_human**, stage lands **awaiting_review** with error banner (never crashes, never silently retries forever).
- Determinism: same patch fixture set + temp 0 → identical aggregate on repeat run (if endpoint proves non-deterministic at temp 0, record and pin **seed** param if available; else accept per-patch variance — the voting layer damps it — and document measured run-to-run grade stability over 10 repeats).
- **Invariant test (the v3 guard)**: for every persisted grading row, **grade == f(tubule_score, pleo_score, mitotic_score)** — enforced as a DB CHECK constraint too:

```
CHECK (grade = CASE WHEN tubule_score+pleo_score+mitotic_score <= 5 THEN 1
```

```
WHEN tubule_score+pleo_score+mitotic_score <= 7 THEN 2  
  
ELSE 3 END)
```

8. Acceptance checklist

- ☐ Full case run on 2 archive slides ≤ 4 min; sub-scores plausible vs original sign-out.
- ☐ Override pleomorphism 2 $\rightarrow$ 3 $\rightarrow$ sum + grade update instantly; chip visible; audit row has justification.
- ☐ Type confirmation gate actually blocks (confirm button disabled until acted on).
- ☐ Malformed-response path lands in **needs_human** gracefully.
- ☐ Grade CHECK constraint deployed; attempt to insert inconsistent row fails.
- ☐ 10-repeat stability run documented (grade identical $\geq 9/10$; else escalate before v4.5).